

Interpreting AI-Based EO Results and Value

Perspectives on AI, Business and Sustainability (PAIBS)

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Term 1 (October 8th - December 3rd) → *From challenge to solution concept*

Definition and Solution Design - identifying meaningful challenges, understanding stakeholder needs, and proposing comprehensive solution.

Term 2 (February 12nd - May 14th) → *From concept to working prototype*

Technical Implementation - technical development, culminating in a fully developed prototype and business case.

Term 2		
Date	Content	Teacher/s
February 12, 2026	Literature Replication + Recap	Marc Herrera
February 19, 2026	Prototyping tools and techniques	Jose Rodriguez
March 12, 2026	Front-end prototyping	Jose Rodriguez
April 9, 2026	Value Experimentation and Validation	Angelo Romasanta
April 16, 2026	Interpreting AI-Based EO Results and Value	Marc Herrera
April 30, 2026	Impact assessment	Angelo Romasanta
May 14, 2026	Final Presentations	Marc Herrera / Angelo Romasanta

Introduction: Grading & Final Deliverables

Task	Grade Weight	Description / Sub-grading
Group project work & presentations	60%	- Problem definition & research quality (20%) - Technical implementation (50%) - Business validation (20%) - Communication & presentation (10%). Individual grade adjusted by peer-review multiplier (x0.8–x1.2).
Midterm individual deliverable	15%	Interview report (user research) and technical literature critique related to the group project.
Final exam	25%	Individual exam covering technical and methodological content from both terms.

Introduction: Group Project Work - 54%

Problem definition & research quality

- Mid-Term Presentation → 0. Problem - Idea - Solution
- 1. Specific, 2. real problem
- 3. Research depth
- 4. User/Client feedback

Technical implementation

- Data and tech used. What do you do and why you do it?
- Results, does it solve the problem? Does it work as intended?
- Quality, code and implementation: organisation, efficiency/optimization, version control
- Basic functionality vs Extended features

Business validation

- Competitors
- Value, benefits. Real validation?
- Feasibility & viability - Costs/Revenue

How innovative is your idea. What do you offer/provide? Does your solution already exist? Costs? Technical complexity? Proof-of-concept now what?

Status - Where are we now, and what have we achieved?

- Current Stage (choose one): Data Preparation / Processing / Model Training / Interpretation
- What is already done (1–2 concrete achievements)
- What is the immediate next step

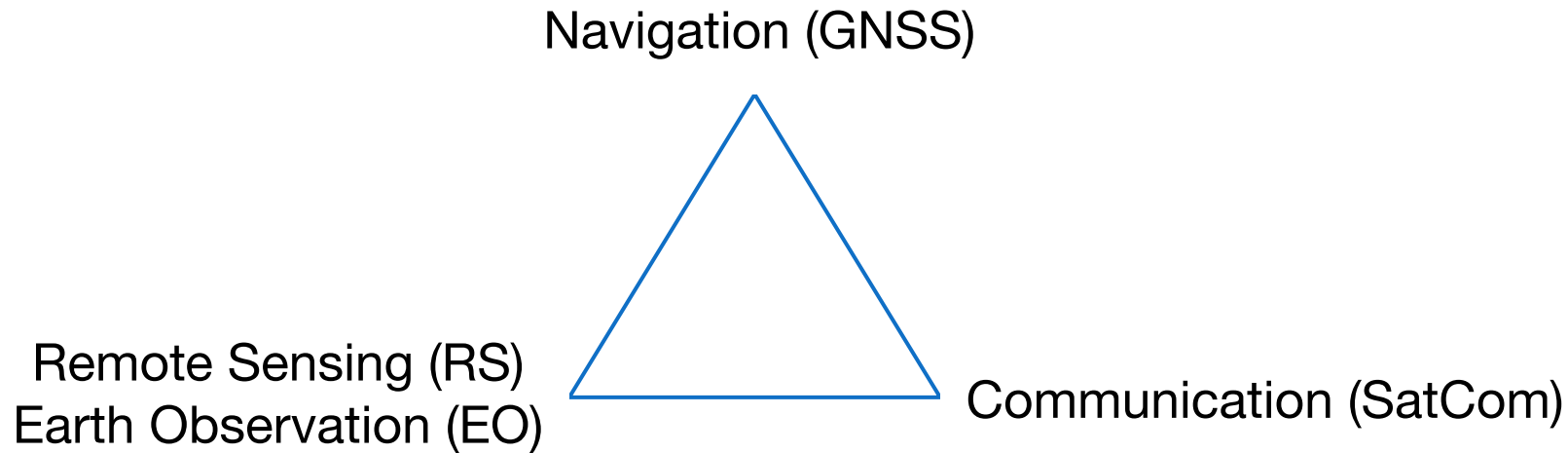
Main Problems - What problems are we facing? Main Problems

- Biggest current problem (only ONE): Data Obtention, Model Performance, Insights-Interpretation
- Why it matters? Impact of the results/project
- What we have tried

Questions - What doubts do we have? One Question

- Technical (e.g., data, model performance), OR
- Conceptual (interpretation, methodology, real value)

→ **3-minute project pitch** (up to 3 slides) - Use the information in the Miro board.



- The space sector is no longer only **governments** → **commercial players** now drive innovation.
- Global market size in **2023** exceeded **\$500 billion**, projected to surpass **\$1 trillion by 2040** (Morgan Stanley / PwC).
- "**NewSpace**": a term for new commercial initiatives — think SpaceX, Planet, Rocket Lab, Starlink.
- Beyond satellites: includes launch providers, ground stations, data analytics companies, AI-driven startups.

Space = infrastructure + services, like roads or internet → enabling countless industries.

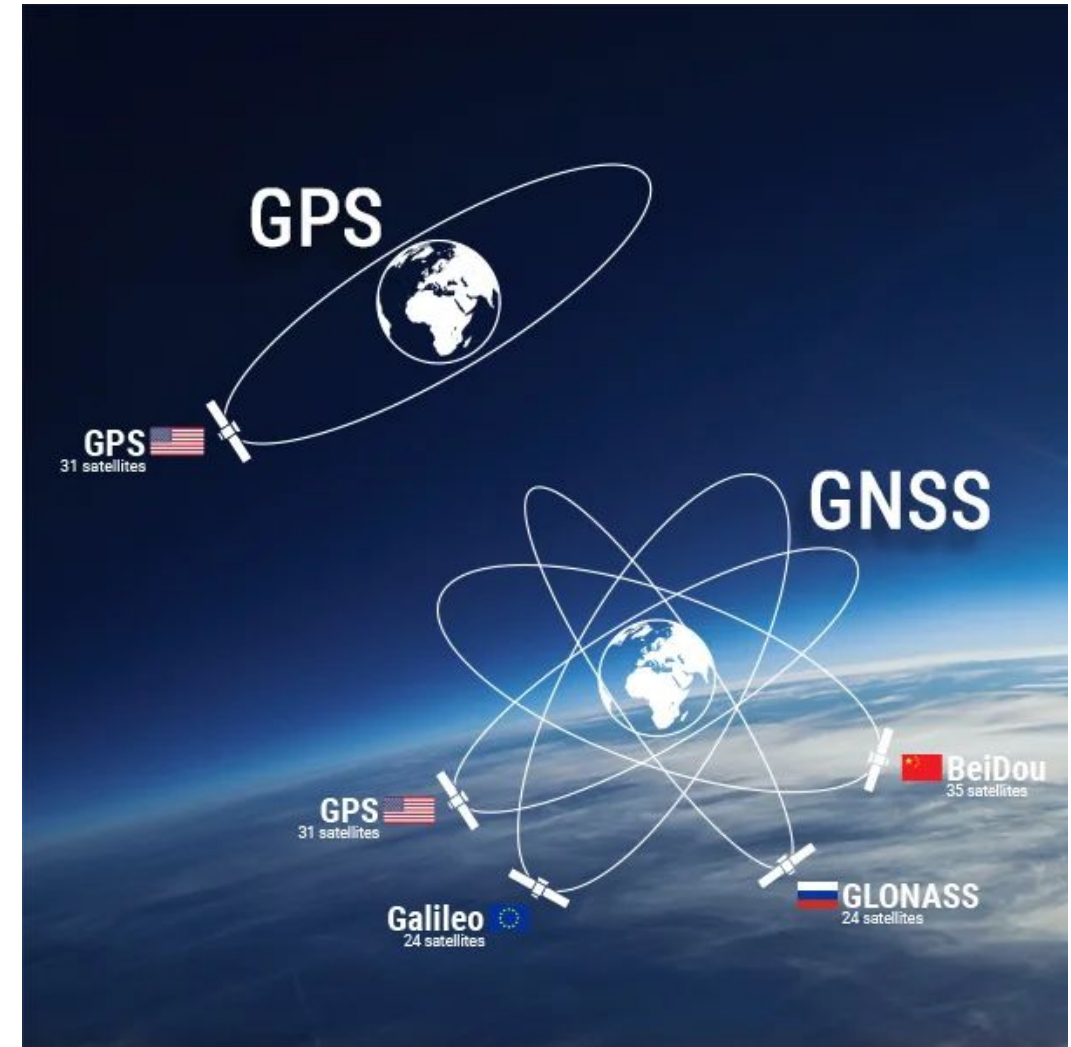
GNSS, *Global Navigation Satellite Systems*, underpin transport & logistics.

Main systems:

- GPS (US, ~1993 full ops)
- Galileo (EU, newer, more precise)
- BeiDou (China, full since 2020)
- GLONASS (Russia)

Uses:

- Finance: stock exchanges use GNSS timing for transactions (microsecond synchronization).
- Everyday life: ride-hailing (Uber), smartphone maps, fitness trackers.
- Emerging uses: drones, precision farming, autonomous vehicles, IoT sensors.
- Business aspect: Galileo is the EU's first major infrastructure project → sovereignty over navigation services.



The Economic Challenge:

Traditional GNSS like GPS or Galileo are "free-to-air" public utilities. Governments provide these signals for free to stimulate global trade, private companies historically struggled to find a business model idea.

→ Navigation as a Service (NaaS) - From Navigation to Resilient PNT (Positioning, Navigation, and Timing).

Problem: GPS signals are weak and easily jammed or "spoofed" (faked).

Solution: Private LEO constellations orbiting 20x closer to Earth. Providing signals 1,000x stronger than GPS, that can penetrate buildings, and are difficult to jam.

Market: autonomous vehicle fleets, 5G networks operators, and high-frequency trading floors. Expected to grow from €73 billion in 2021 to €250 billion by 2031.

Founded by Stanford engineers, Xona is building Pulsar, the first private LEO navigation constellation. High-performance "upgrade" to GPS.

LEO signal is 100x–1,000x stronger than GPS, allowing it to work indoors, in dense urban areas and more securely.

Financial Evolution & Business Milestones:

- 2019–2021 (R&D): Focused entirely on software-defined radio and signal design. They survived on early-stage grants and seed funding with no revenue.
- 2022 (Proof of Concept): Launched Huginn, the first-ever commercially funded navigation satellite.
- 2025–2026 (Commercial Scaling): "High Burn" phase. Funding invested to mass-produce satellites for a full constellation.



Funding: Raised a **\$92M Series B** in 2025 followed by a **\$170M Series C** in April 2026 to mass-produce their 300-satellite constellation.

Revenue Strategy: Subscription-based or "lifetime fee" models for industrial users in agriculture, defense, and autonomous transport.

GPS = "Free Public WiFi," Xona = "Private 5G Network" - higher cost, but reliable.

"Big Fish" - "Small Fish"

Satelles proved the business case and was acquired by a major player.

1. Satelles developed "Satellite Time and Location" (STL) using the existing Iridium satellite network. It provides a secure backup that works indoors.
2. Iridium acquired the remaining 80% of Satelles for \$115 million in early 2024.

Value

They sell to 5G base stations and data centers that require "unhackable" timing to stay synchronized.

They expect this new navigation business line to generate over \$100 million in annual service revenue by 2030.

Traditional SatCom: TV & radio broadcasting, military communications. Later regular comms.

Now:

- Mega-constellations (e.g., Starlink with >6,000 satellites)
→ Global **internet**: planes, cruise ships, rural areas.
- SatCom critical in disaster relief when ground networks fail (earthquakes, floods).
- Maritime & aviation: allows real-time data, fuel optimization, safety tracking.

Impact:

Business: SatCom enables inclusion → connecting billions without internet.

Geopolitics: broadband constellations influence digital sovereignty.

From broadcast-only to two-way, low-latency internet → direct competition with terrestrial providers

Traditional European **GEO** giants are merging and acquiring to compete with **LEO** constellations like Starlink.

SES (Luxembourg) Recently acquired Intelsat for roughly \$3.1 billion, aiming to create a "multi-orbit powerhouse".

Eutelsat (France) Merged with OneWeb to combine traditional video broadcasting revenue with high-speed LEO connectivity.

But's

- SES shares dropped 13.5% in late 2025 after post-merger earnings forecasts disappointed investors, largely due to **high debt from the Intelsat** deal (net debt reached €6 billion).
- Traditional video revenue (formerly 61% of Eutelsat's group revenue) is declining as users shift to **streaming**, forcing these companies to pivot to "Connectivity" services, which now represent over 56% of backlogs.

Remote Sensing (RS) / Earth Observation (EO)

Remote Sensing (RS) is the **collection of data** about the Earth's surface without direct contact.

RS allowed us to **monitor and study our planet** in new ways, providing a global view of the Earth and enables observation of changes over time.

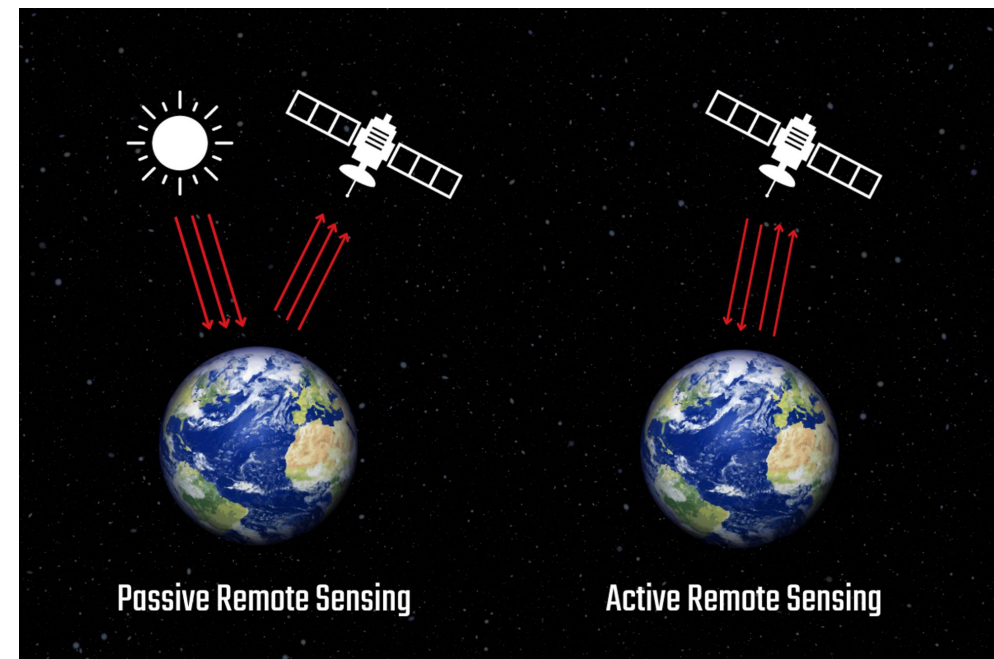
EO **combines data** from multiple sources to create comprehensive views of the planet.

RS provides the data — **EO** turns it into **understanding**.

The EO space sector hosts a wide range of use cases, business models, and emerging companies growing each year.

EO value chain:

- Upstream (hardware)
- Downstream (data = services)



HR Multispectral EO Images - Planet

World's largest constellation of EO satellites (~200). Daily HR optical images and AI-driven insights (NVIDIA partnership) for agriculture, defense, and climate monitoring.

Future Evolution

- The Defense & Intelligence segment saw revenue growth of 70% year-over-year in 3Q26.
- Currently holds a total backlog of approximately \$734.5 million.
- Recently achieved its fourth consecutive quarter of positive adjusted EBITDA (\$5.6M in late 2025) and is generating positive free cash flow.

Economic Viability Evolution:

Planet's strategy relies on "operating leverage". The cost once the satellite is in orbit, to sell that data to a second or hundredth customer is nearly zero. However, the company faces "capital intensity risks" due to the need for frequent constellation refreshes every 3–5 years (+400 satellites launched).



The Future of the Space Sector

Space Exploration (Moon, Mars, beyond)

Renewed race: NASA's Artemis program aims for a permanent Moon base by 2030.

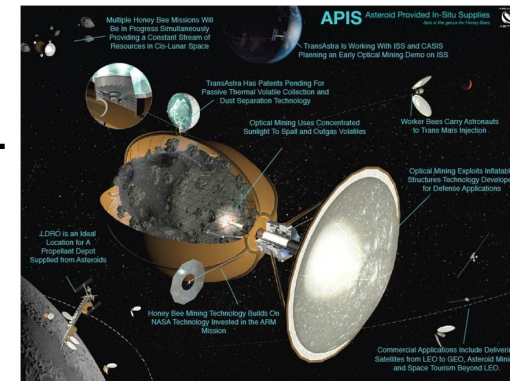
Mars ambitions: SpaceX's Starship designed for interplanetary transport.



Space Mining (asteroids, lunar resources)

Asteroids and the Moon hold vast amounts of metals (platinum, rare earths, water ice).

Potential to fuel rockets with lunar water → reduces cost of deep space missions.



Space Tourism (suborbital, orbital stays)

First wave is suborbital flights (Virgin Galactic, Blue Origin).

Orbital tourism: SpaceX Crew Dragon has already flown private astronauts.

Prices falling (though still >\$200k per seat), opening future markets.



Orbital Habitats (private space stations)

ISS is aging → private replacements on the way.

Axiom Space building commercial station modules.

VAST Space developing inflatable orbital habitats → first private space station around 2025–2030.



Economic Viability

SpaceX has transformed from a **high-risk** startup **into** the **main force** in the global space economy. Its business model is a self-reinforcing "flywheel" where **revenue from launch services funds** the **R&D** for next-generation systems like Starship (also Starlink).

Value

SpaceX produces nearly 70% of its components in-house, reducing costs by an order of magnitude (e.g., \$5,000 for a radio vs. the \$100,000 industry standard).

Failure to Success

The Falcon 1 program nearly bankrupted the company after **three** consecutive launch **failures** between 2006 and 2008. SpaceX had enough capital for only one final attempt; the **fourth** launch's **success** on September 28, 2008.



From a "black hole" for capital to a profit machine. The +20 years road to reach a state of consistent net profitability.

- 2002 – 2008 (Survival Phase): Founded with \$100M of Elon Musk's personal money. Three consecutive launch failures of Falcon 1 Success on the 4th launch in late 2008, securing a \$1.6B NASA contract that prevented bankruptcy.
- 2009 – 2019 (R&D Burn): Massive reinvestment of launch revenue into Falcon 9 reusability and the Dragon spacecraft. While cash flow appeared periodically, **the company remained "net loss" due to billions in capital expenditure (CapEx).**
- 2020 – 2023 (Starlink Pivot): Capital needs spiked as SpaceX began launching thousands of satellites. Starlink was a massive cash drain until it reached a critical mass of subscribers.
- 2024 – 2026 (Profit Era):

2024: **First**-ever full **year** of Starlink **profitability** (\$72.7M net profit).

2025: Generated an estimated \$8 billion in EBITDA on ~\$16B revenue.

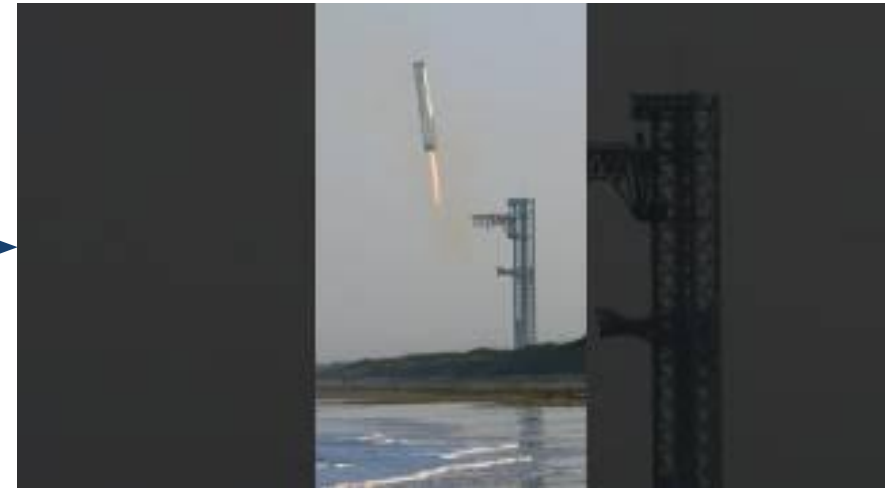
2026: Preparing for a historic IPO with a valuation target above \$2 trillion.

SpaceX Business History

2008



2025



What? → **Luxury tourism**, a fragile market compared with SpaceX "utility" (internet/cargo). **IEEC[®]**

However → Only \$1.54 million revenue in FY2025, down 78% from the previous year. With sustained an operating loss of \$285.3 million in 2025.

Investors grew tired of "protracted cash burn" and repeated flight delays = Stock price crashed by over 85% between 2021 and 2025 (\$100M net loss per quarter) as

The Problem

The model required hundreds of flights per year to break even.

- Technical issues.
- Lack of Interest.
- High associated costs.

Todays reality, the company is in "survival mode." The stock has lost 90% of its value. *Space tourism without high frequency is a "cash incinerator."*



Orbital Real Estate & Microgravity Manufacturing

Axiom: Orbital real estate. They are building the **first commercial space station** to replace the ISS.

- Raised over \$700M (Series C) but operating at a heavy loss. "CapEx-heavy" model, spend billions manufacturing modules before they can generate high-margin rent.
- Revenue will come from "Sovereign Astronaut" missions (governments paying for research) and a \$2.2B NASA contract for lunar suits and station modules.

In 2025, their valuation dipped from \$2.6B to \$2B due to module delays. Investors punish "future ideas" when timelines slip.

Varda: Microgravity to create physically impossible things (pharmaceuticals, fiber optics, semiconductors) on Earth.

- Raised \$187M through 2025. All capital is going into R&D for reentry capsules. Their 2024 success in growing HIV drug crystals was a "proof of concept," not a commercial revenue stream.

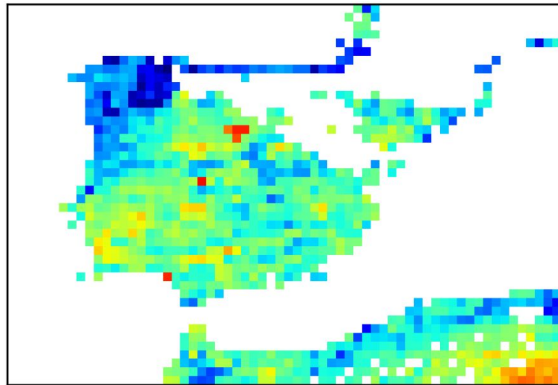
Investors are buying a "call option" on the future of medicine. If Varda can't solve the high cost of "shipping" materials back to Earth, the business model collapses.

Examples of ROI = 0

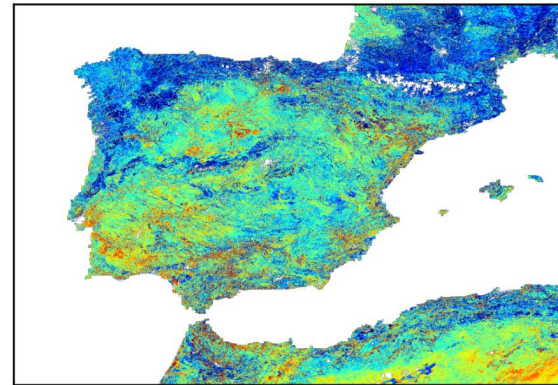


Soil Moisture (SM) Super Resolution

Original CCI SM (25 km)



Estimated SM (1 km)

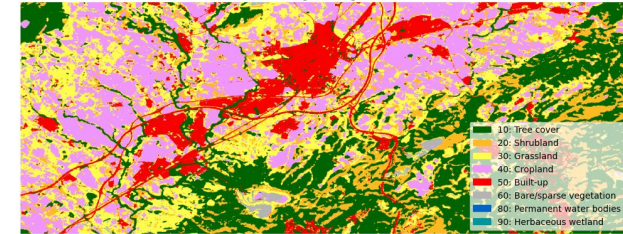


Land Cover (LC) Classification

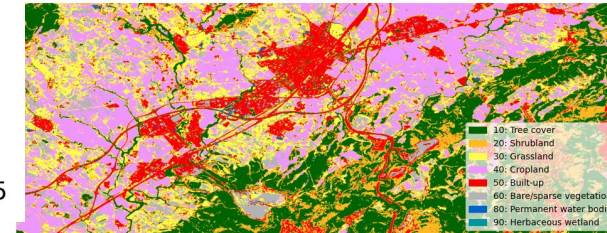
Optical RGB image



Original image



Model Classification



Activity

How would you start?

1. Value Proposition

- What problem does this solve?
- What is the *practical benefit*?

2. Customer - Who would actually pay for this to whom is this solving a problem?

(Be specific: company, sector, role)

3. Monetization

- How do we make money? Subscription? API? Reports? SaaS?
- Give one concrete pricing example (e.g., €/month, €/use, contract)

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Estimate:

- Number of clients (Year 1 → Year 3)
- Basic revenue evolution.

When do you reach **break-even (approx.)**?

4. Go-to-Market (very simple)

- How would you start? Pilot? Partnership? Open demo?